Feedforward Neural Network Quiz

Consider a simple feedforward neural network with 4 input nodes, 2 hidden layers each with 4 nodes, and 3 output nodes. The architecture is as follows:

- **Input Layer:** 4 nodes (I1, I2, I3, I4)
- **Hidden Layer 1:** 4 nodes (H1, H2, H3, H4)
- **Hidden Layer 2:** 4 nodes (H5, H6, H7, H8)
- **Output Layer:** 3 nodes (O1, O2, O3)

1. Draw the complete feedforward neural network diagram.
2. Label each node clearly as shown.
3. Write the mathematical notations for each connection above the arrows, where:
    - **W1** represents the weight matrix between the input and hidden layer 1.
    - **W2** represents the weight matrix between hidden layer 1 and hidden layer 2.
    - **W3** represents the weight matrix between hidden layer 2 and the output layer.
4. Indicate that the activation function used in each layer is **Sigmoid** for hidden layers and **Softmax** for the output layer.

Hint:

- Each arrow from one node to another represents a connection with a weight that is a part of the corresponding weight matrix (W1, W2, W3).
- Use the following notations for each layer:

In a simple feed forward neural network with 4 input nodes, 2 hidden layers each with 4 nodes, and 3 output nodes, what is the total number of parameters (weights and biases) in the network?

- a) 40
- b) 38
- c) 45
- d) 42

Given the following architecture:
- Input layer: 4 nodes
- Hidden layer 1: 4 nodes
- Hidden layer 2: 4 nodes
- Output layer: 3 nodes

What is the equation for the transformation from the input layer to the first hidden layer, assuming a weight matrix W_1 and bias vector b_1?

- a) $h_1 = f(W_1 \cdot X + b_1)$, where $W_1 \in R^{(4 \times 4)}$ and $b_1 \in R^{(4)}$
- b) $h_1 = f(W_1 \cdot X + b_1)$, where $W_1 \in R^{(4 \times 5)}$ and $b_1 \in R^{(4)}$
- c) $h_1 = f(W_1 \cdot X + b_1)$, where $W_1 \in R^{(5 \times 4)}$ and $b_1 \in R^{(4)}$
- d) $h_1 = f(W_1 \cdot X + b_1)$, where $W_1 \in R^{(4 \times 3)}$ and $b_1 \in R^{(3)}$

In a feedforward neural network, if the activation function for the hidden layers is ReLU (Rectified Linear Unit), what is the equation that describes the activation function for each hidden unit?

- a) $f(z) = \max(0, z)$
- b) $f(z) = 1 / (1 + e^{(-z)})$
- c) $f(z) = z^2$
- d) $f(z) = z$

In a feedforward neural network with 2 hidden layers, where the first hidden layer has 4 nodes and the second hidden layer has 4 nodes, how many weight matrices are there, and what are their dimensions?

- a) 3 weight matrices: $W_1 \in R^{(4 \times 4)}$, $W_2 \in R^{(4 \times 4)}$, $W_3 \in R^{(3 \times 4)}$
- b) 3 weight matrices: $W_1 \in R^{(4 \times 4)}$, $W_2 \in R^{(4 \times 4)}$, $W_3 \in R^{(4 \times 3)}$
- c) 3 weight matrices: $W_1 \in R^{(4 \times 4)}$, $W_2 \in R^{(3 \times 4)}$, $W_3 \in R^{(4 \times 3)}$
- d) 3 weight matrices: $W_1 \in R^{(3 \times 4)}$, $W_2 \in R^{(4 \times 4)}$, $W_3 \in R^{(4 \times 3)}$

Which of the following describes the process of feedforward in a neural network?

- a) Input data is passed through the network layer by layer, and the final output is computed from the last layer.
- b) The output of each node is calculated by summing the weighted inputs, passing through the activation function, and then passed on to the next layer.
- c) Both a and b
- d) None of the above

Consider a neural network with 4 input nodes, 2 hidden layers each with 4 nodes, and 3 output nodes. If the activation function is Sigmoid in both hidden layers and the output layer, which of the following is the correct transformation from the second hidden layer to the output layer?

- a) $y = \sigma(W_3 \cdot h_2 + b_3)$, where $\sigma(z) = 1 / (1 + e^{(-z)})$
- b) $y = \sigma(W_2 \cdot h_2 + b_2)$, where $\sigma(z) = 1 / (1 + e^{(-z)})$
- c) $y = \sigma(W_1 \cdot h_2 + b_1)$, where $\sigma(z) = 1 / (1 + e^{(-z)})$
- d) $y = \sigma(W_1 \cdot h_1 + b_1)$, where $\sigma(z) = 1 / (1 + e^{(-z)})$

What is the primary purpose of using an activation function in each neuron in a feedforward neural network?

- a) To introduce non-linearity, allowing the network to model complex functions
- b) To scale the input values to a fixed range
- c) To normalize the outputs of the neurons
- d) To prevent overfitting

Given the following network architecture:
- Input layer: 4 nodes
- Hidden layer 1: 4 nodes
- Hidden layer 2: 4 nodes
- Output layer: 3 nodes

If the activation function is tanh for the hidden layers and softmax for the output layer, which of the following represents the output layer equation?

- a) $y = \text{softmax}(W_3 \cdot h_2 + b_3)$
- b) $y = \text{softmax}(W_2 \cdot h_2 + b_2)$
- c) $y = \text{softmax}(W_1 \cdot h_1 + b_1)$
- d) $y = \text{softmax}(W_3 \cdot h_2 + b_2)$

Which of the following is true regarding the output layer in a feedforward neural network for a multi-class classification task?

- a) It uses a softmax activation function to ensure the output probabilities sum to 1.
- b) It uses a ReLU activation function to produce the output values.
- c) It uses a Sigmoid activation function for multi-class classification.
- d) It has no activation function, as the raw output values are sufficient for classification.

In a feedforward network with 4 input nodes, 2 hidden layers with 4 nodes each, and 3 output nodes, how many total biases are there in the network?

- a) 11
- b) 12
- c) 13
- d) 14